



02.07.2026
Tag 9

Azure for Students

Learning Path
Networks

→ ^{meine} Subscription 100\$ 1 Jahr
Von KI erstellt, von Menschen kontrolliert.

Part 1 ✓

Part 2 ✓

Part 3 ✓

move

Mein Tenant

Der Dritte Mann

GitHub Repo <https://github.com/www42/a5>

Part 4

AZ-104

Learning Paths:

- Manage identities and governance in Azure
- Prerequisites for Azure administrators – CloudShell and templates
- Configure and manage virtual networks for Azure administrators
- Implement and manage storage in Azure
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

What happened yesterday



Azure Copilot (kostenlos!)

Gemini
Gemini

GPT 5.5
Fables

What are agents?

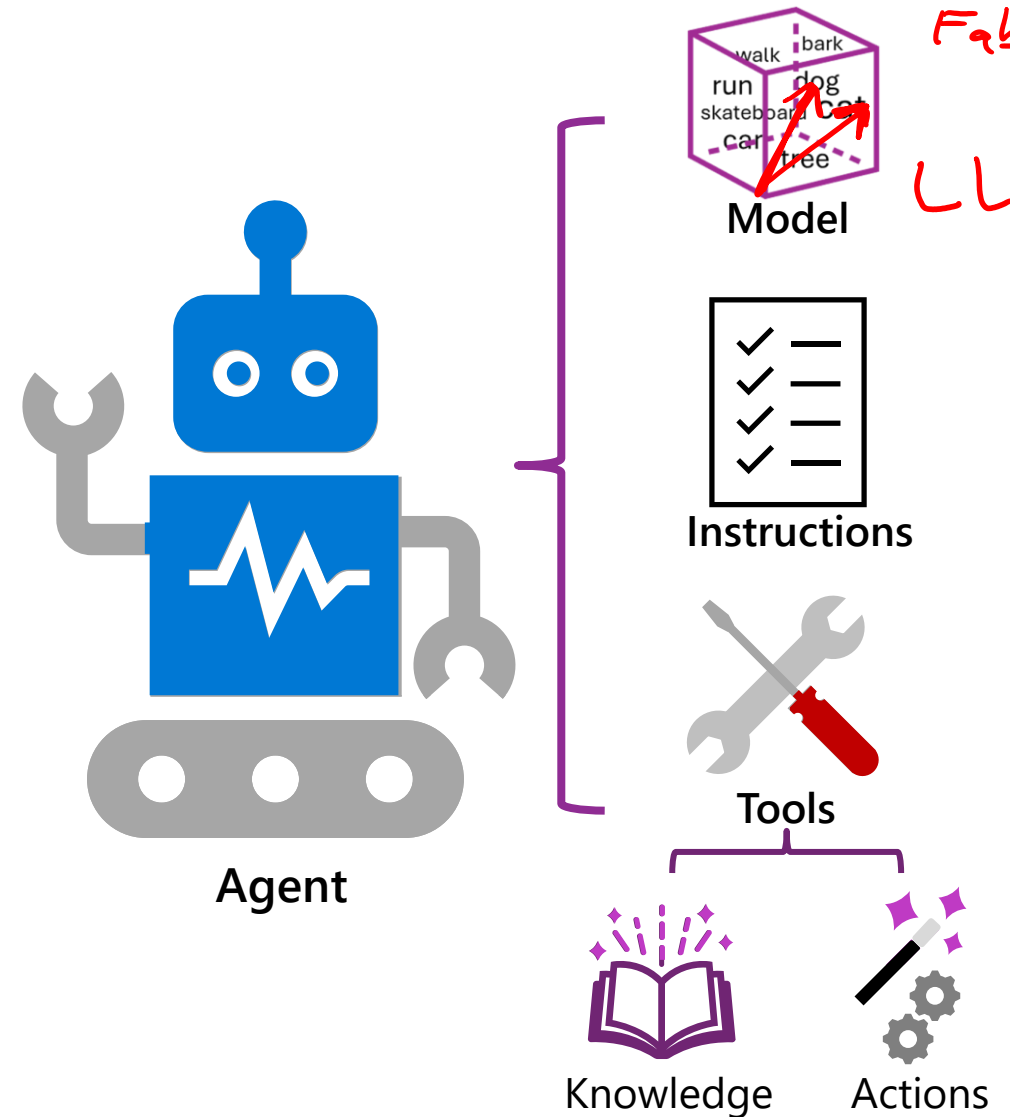
Autonomous, action-capable AI assistants

Agents use:

- A **model** for language understanding and reasoning
- **Instructions** to define behavior and scope
- **Tools** they can use to:
 - Find **knowledge** to contextualize their work
 - Perform **actions** to automate tasks

Agents collaborate with:

- Human users, to assist with tasks and processes
- Other agents, to create workflows of specialized agents, each with their own responsibilities



Part 1

GfN Tenant



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GfN GmbH (Edu)



Subscription 100\$

Part 2

Mein Tenant



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atropos69123.oumicrosoft.com

No Subscription

move

Azure for Students

Schicksal göttinnen

Atropos

Lachesis

Klotho

schneidet ab

Länge

spinnt

AZ-104

 Configure and manage virtual networks for Azure administrators

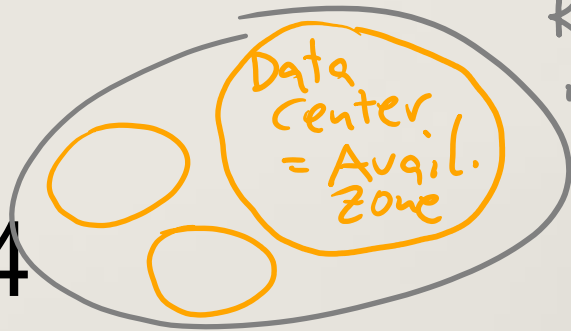




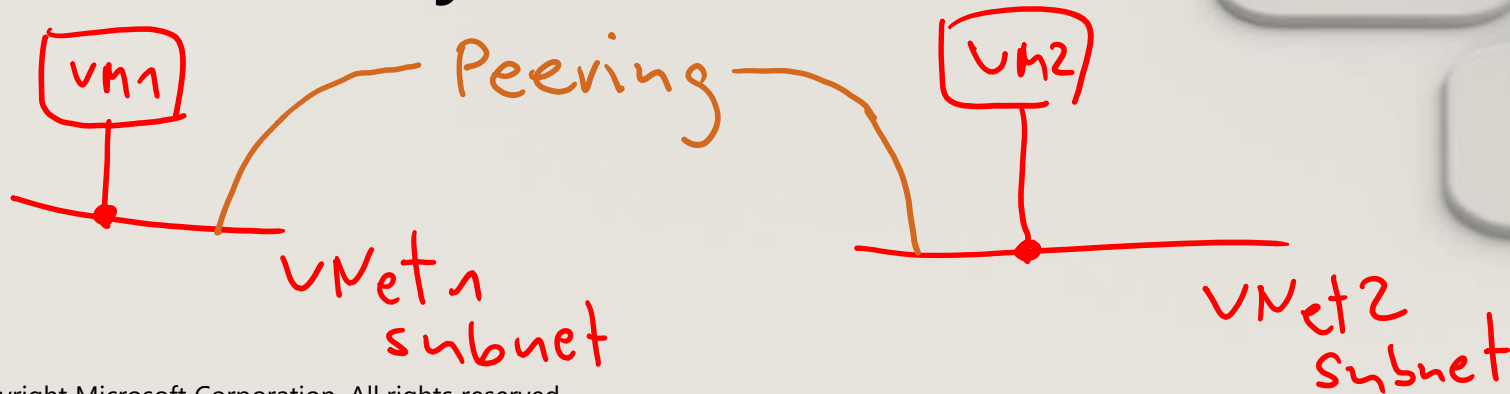
SDN Software Defined Networking

Hub-Spoke
Topology

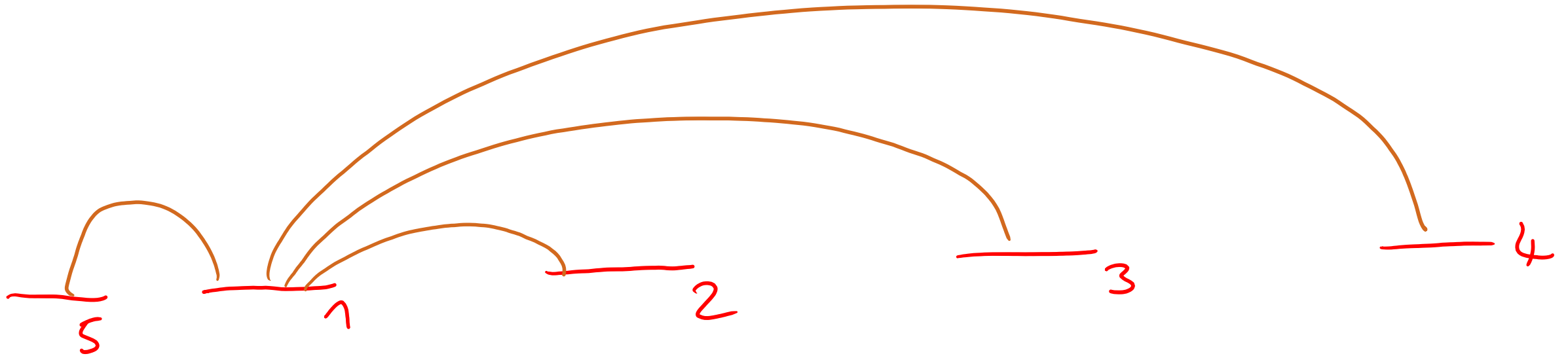
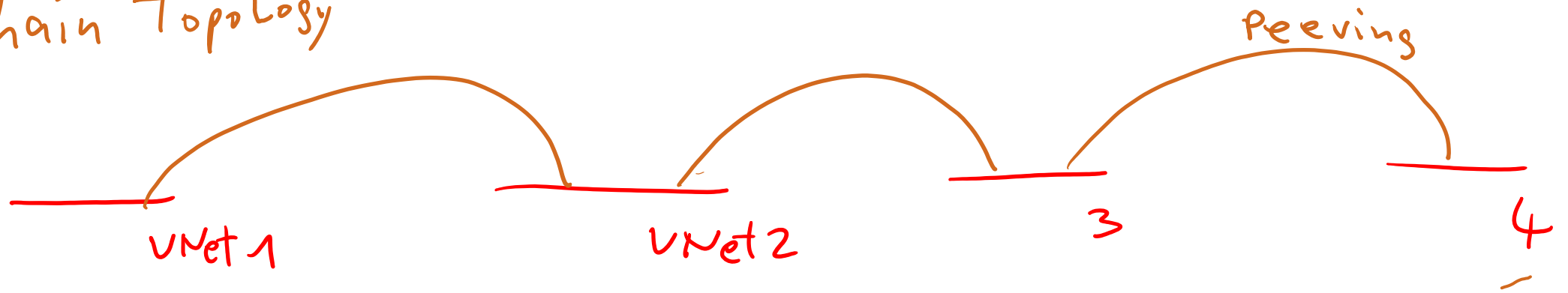
AZ-104



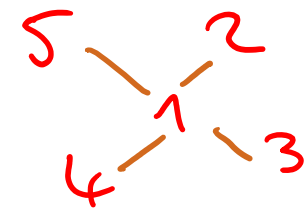
Administer Intersite Connectivity



Chain Topology



Hub-Spoke Topology

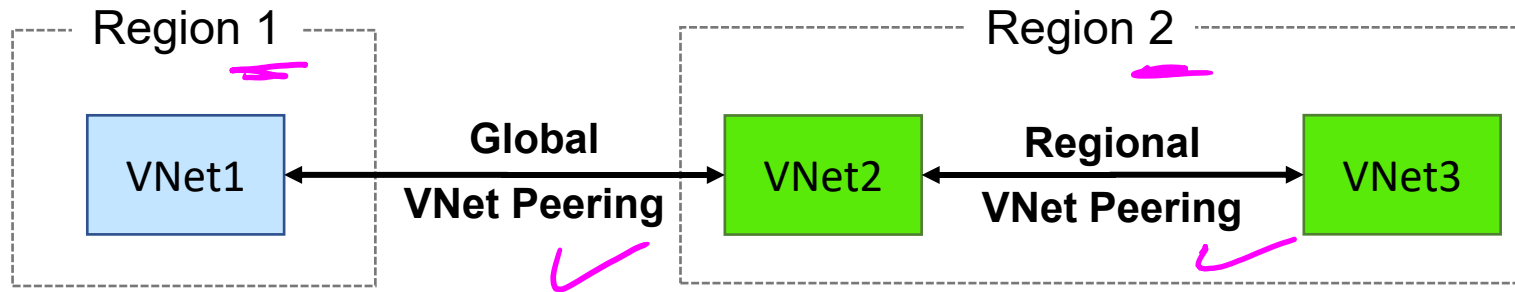


Configure VNet Peering



Determine VNet Peering Uses

Peering is not encrypted.
Encryption? → VPN
(Virtual Network Gateway)



- Two types of peering: Global and Regional
- Connects two Azure virtual networks – you can peer across subscriptions and tenants
- Peered networks use the Azure backbone for privacy and isolation
- Easy to setup, seamless data transfer, and great performance

GW = Default Gateway

BGP dynamic Routing

Determine Gateway Transit and Connectivity Needs

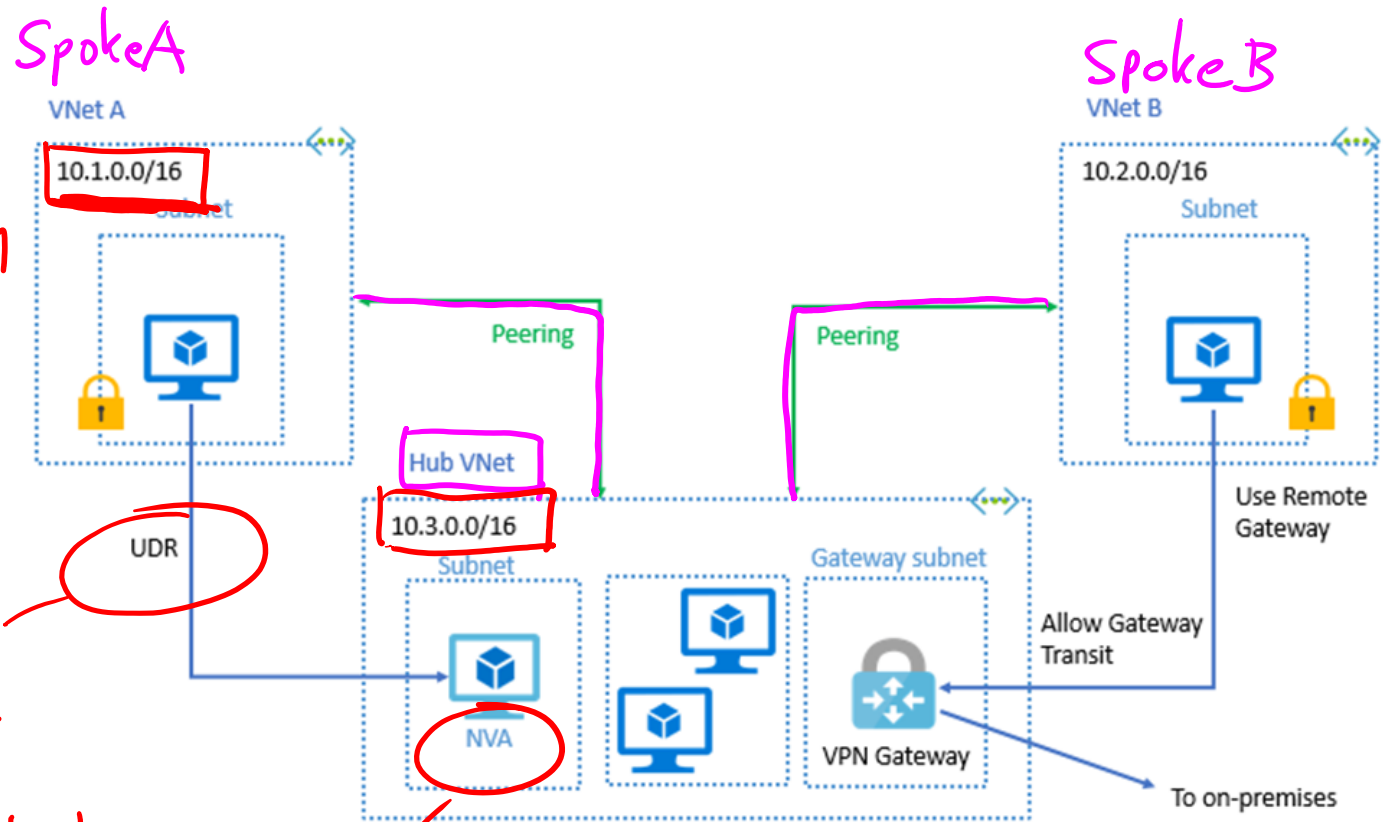
- Gateway transit allows peered virtual networks to share the gateway and get access to resources
- No VPN gateway is required in the peered spoke virtual network
- Default VNet peering provides full connectivity

10.1.0.1
Default GW

Static Routing

User Defined Route

Network Virtual Appliance



IP address spaces of connected networks can't overlap

Create VNet Peering

- Allow virtual network traffic
- Allow forwarded traffic
- Allow gateway or route server traffic
- Status should show "connected"

Remote virtual network summary

Peering link name *

Remote virtual network peering settings

Allow 'vnet-2' to access 'vnet-1' ⓘ ☒

Allow 'vnet-2' to receive forwarded traffic from 'vnet-1' ⓘ ☒

Allow gateway or route server in 'vnet-2' to forward traffic to 'vnet-1' ⓘ ☐

Enable 'vnet-2' to use 'vnet-1's' remote gateway or route server ⓘ ☐

Local virtual network summary

Peering link name *

Local virtual network peering settings

Allow 'vnet-1' to access 'vnet-2' ⓘ ☒

Allow 'vnet-1' to receive forwarded traffic from 'vnet-2' ⓘ ☒

Allow gateway or route server in 'vnet-1' to forward traffic to 'vnet-2' ⓘ ☐

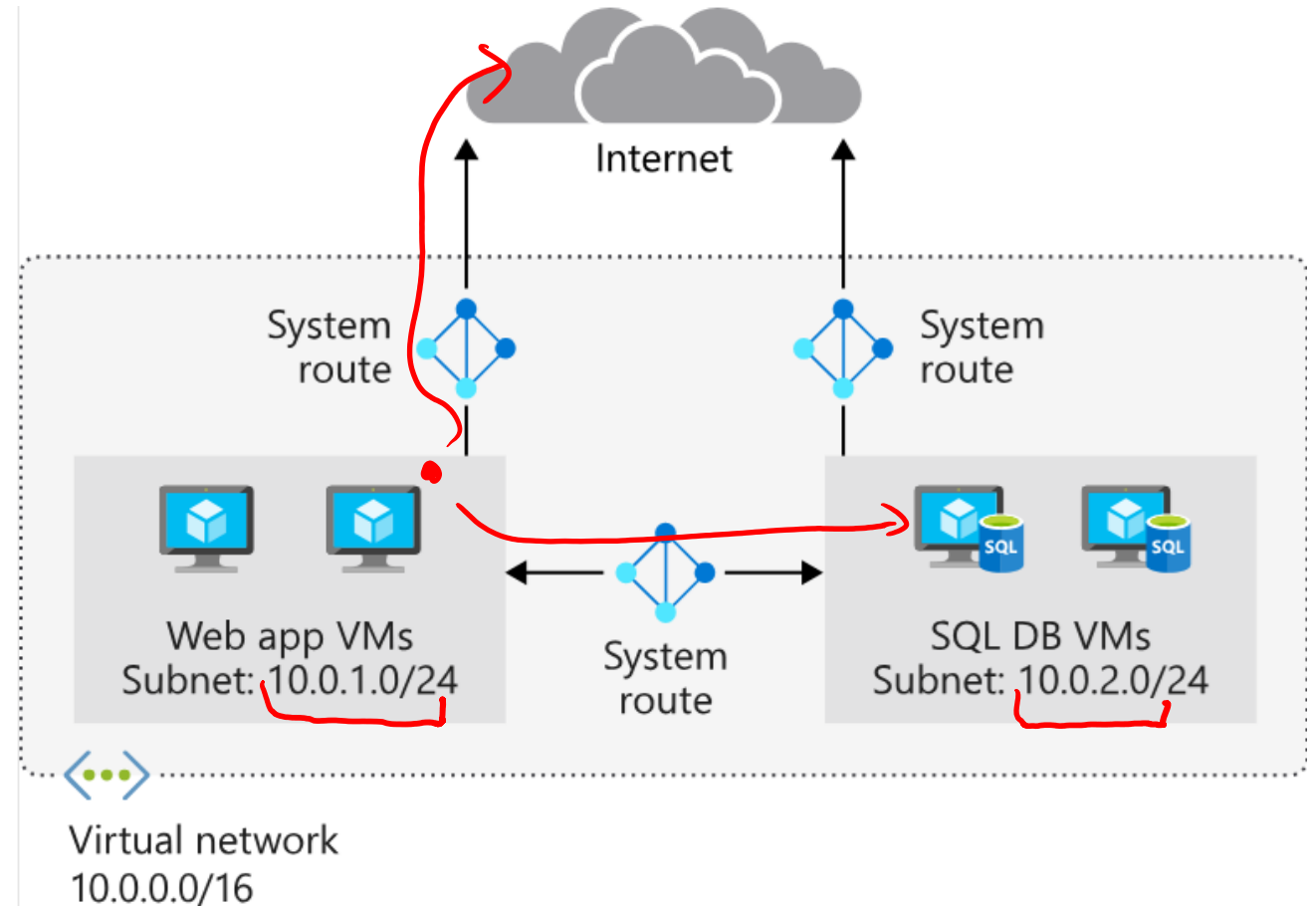
Enable 'vnet-1' to use 'vnet-2's' remote gateway or route server ⓘ ☐

Manage and control traffic flow with routes



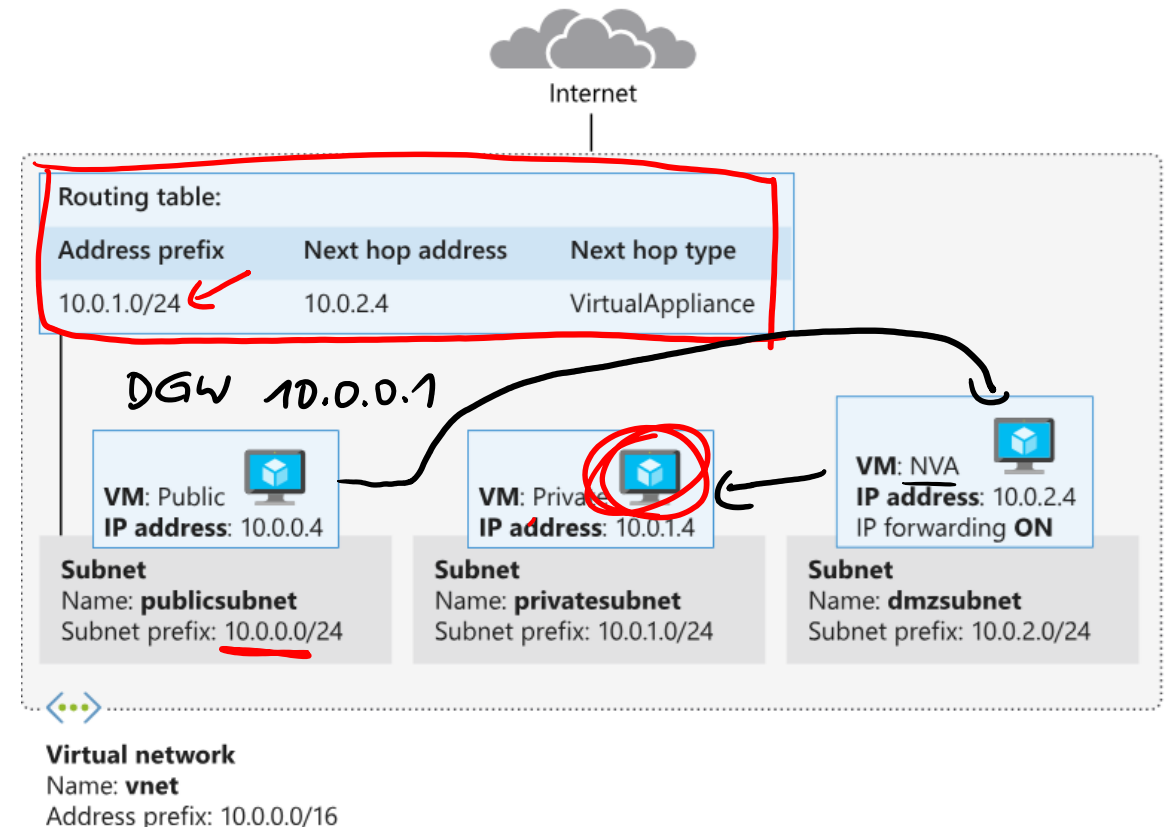
What are System Routes?

- Pre-defined routes automatically added to every subnet's route table by Azure
- Ensures fundamental connectivity (e.g., VNet-to-VNet, VNet-to-Internet, VNet-to-Azure services)
- **Examples:** Routes for VNet peering, local traffic, and the 0.0.0.0/0 internet route
- You can't delete system routes, but you *can* override with custom routes (UDRs)



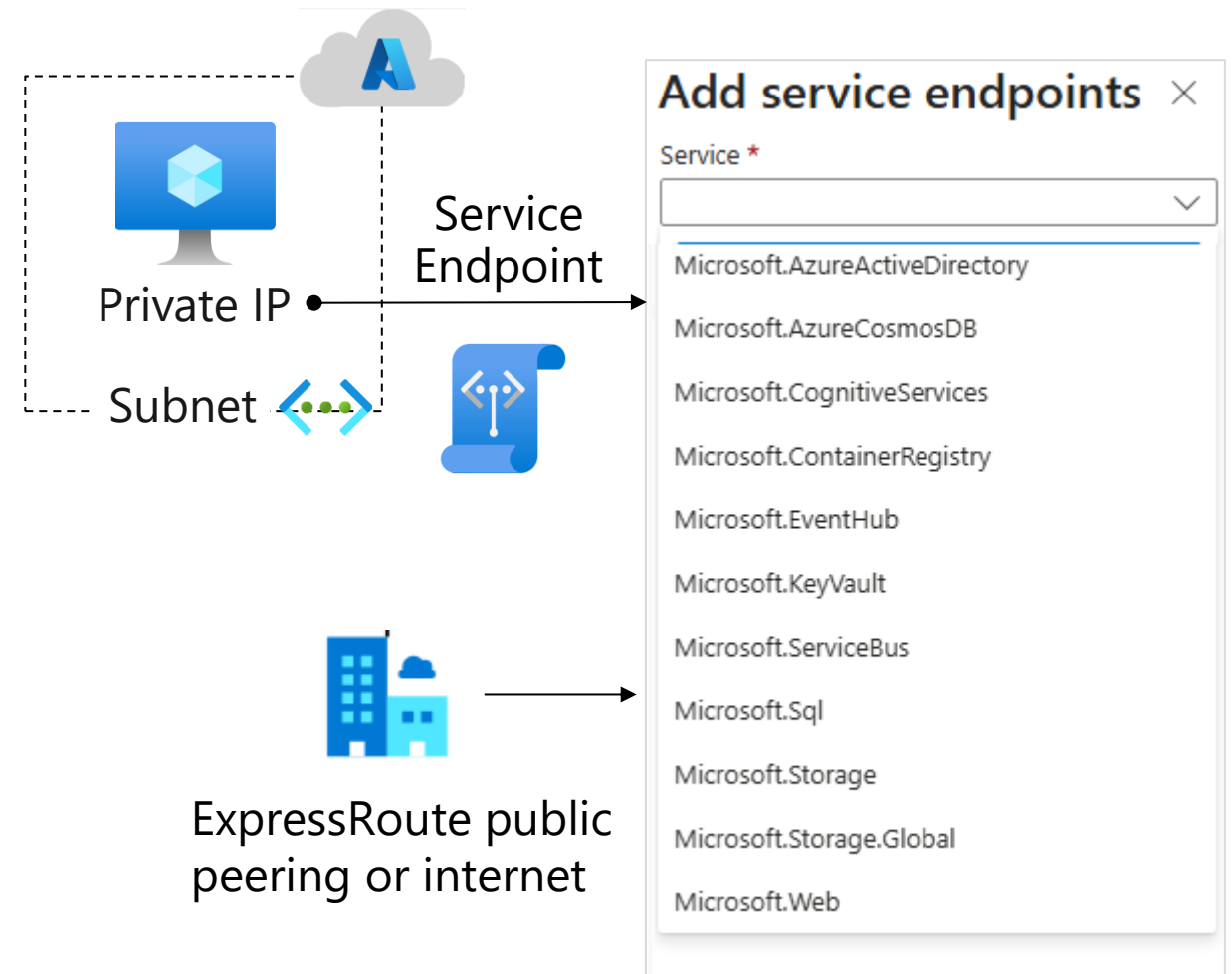
What are User-Defined Routes (UDRs)?

- UDRs override the default system routes
- Traffic can be directed to virtual appliance, Virtual Network Gateway, Virtual Network, Internet, or None (drop it)
- Border Gateway Protocol (BGP) exchanges routes between Azure and on-premises networks
- You can specify a service tag as the address prefix for a user-defined route instead of an explicit IP range

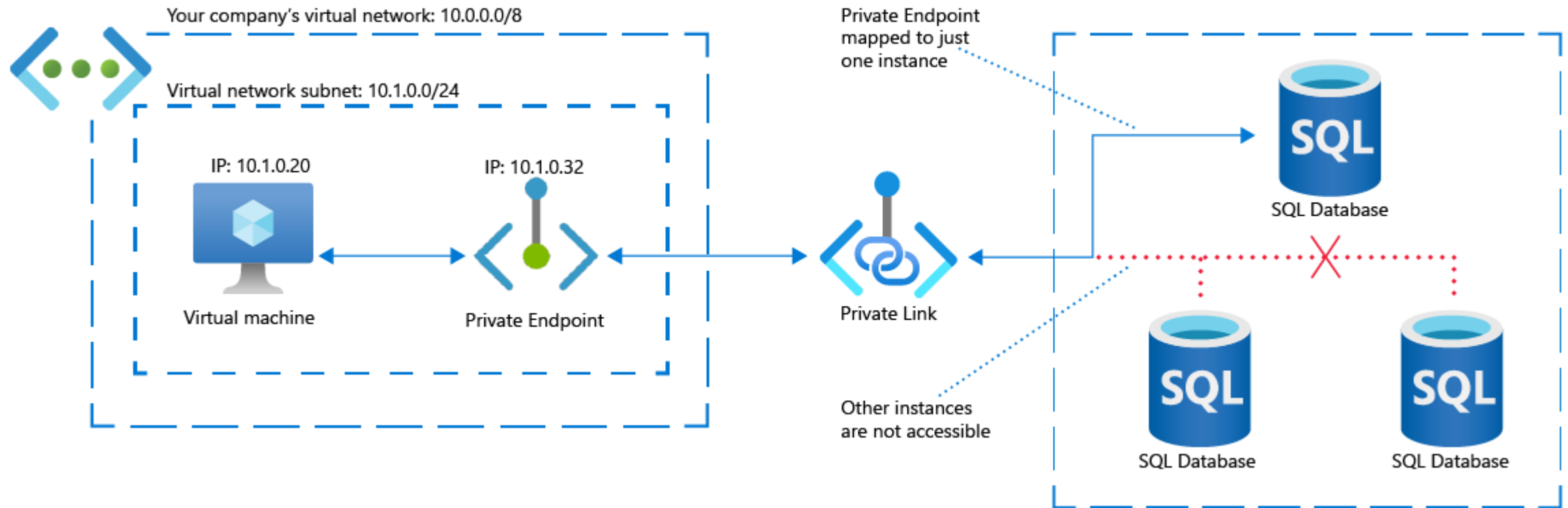


What is a Service Endpoint?

- A direct connection from your virtual network to supported Azure services
- Restricts the flow of traffic and enables your Azure VMs to access the service directly from your private address space
- Traffic always stays on the Microsoft Azure backbone network



What is an Azure Private Endpoint?



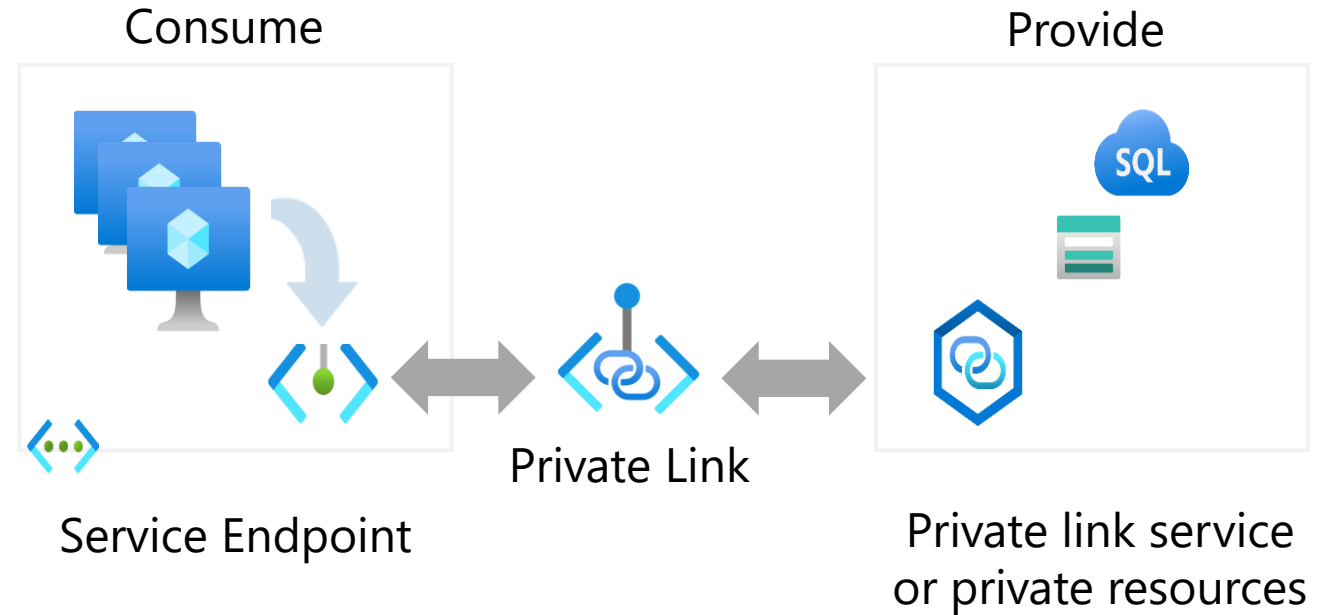
Uses a private IP address from your virtual network

Granular access to services

Eliminates the potential security risk of a public IP address

What is Azure Private Link?

- Build and allow private connections between resources
- To privately connect to a service, create a service endpoint
- To privately render a service, create a private link service or private resources
- All data travels on the Microsoft backend



Compare Service Endpoints and Private Endpoints

Feature	Service Endpoint	Private Endpoint
Use Case	Simple, cost-effective way to secure traffic to Azure services from within VNet	Maximum security, regulatory compliance, hybrid connectivity (on-prem to private IP).
Security	Secures traffic from VNet to service, but the service endpoint remains publicly accessible (can be firewall-restricted).	Highest security; traffic stays within VNet/Microsoft backbone, no public exposure; excellent for data exfiltration prevention.
On-premises access	No direct access (via public IP)	Yes (via VPN/ExpressRoute to private IP)
IP Address	Public IP of service (traffic uses private VNet IP to reach it)	Private IP from your VNet
DNS	No change needed (resolves to public IP)	Requires Private DNS Zone
Cost	Free (no added cost)	Cost for endpoint & data processed

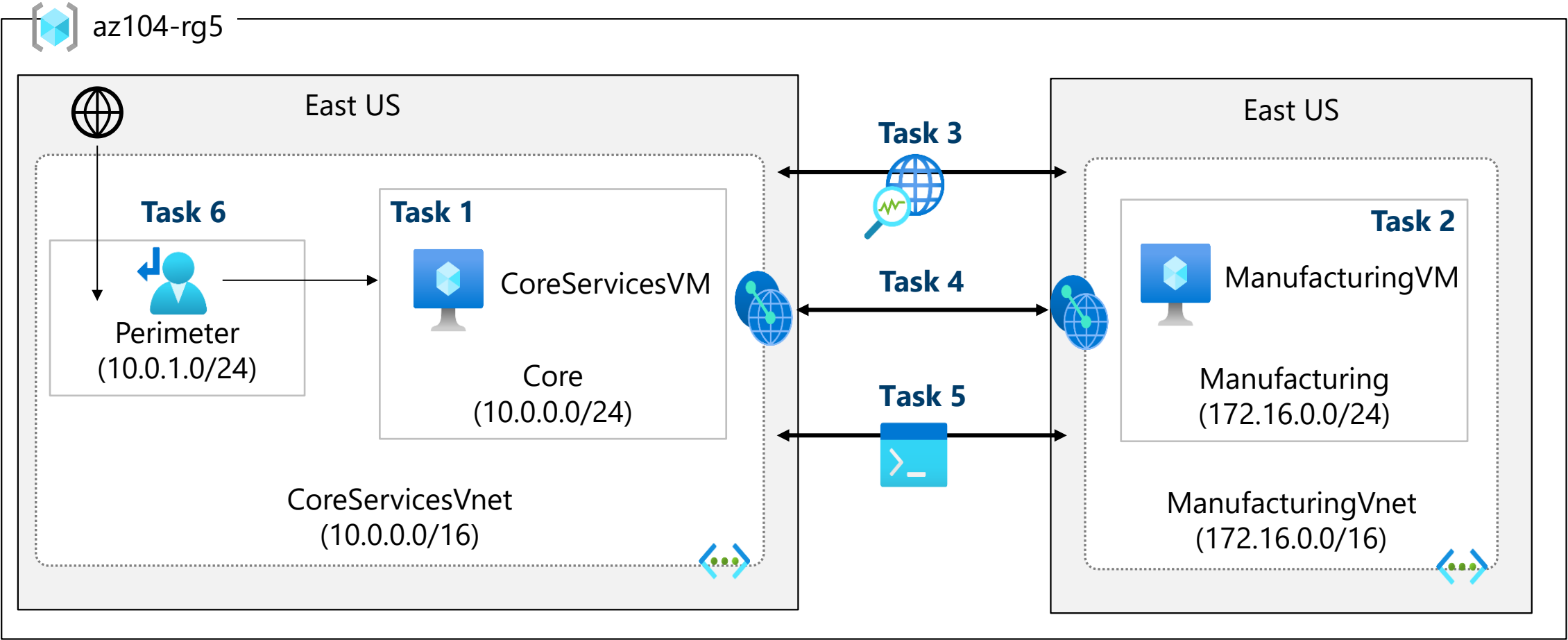
Labs



Lab 05 - Implement Intersite Connectivity



Lab 05 – Architecture diagram



End of presentation



